# TITLE PLACEHOLDER

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**Abstract**

What is a second telling of an experience a telling of? Memories change with time and with retrieval, so each recounting may be shaped less by the experience itself than by how we last recounted it. Here we use a geometric framework to compare two recountings of the same experience. We model an experience and each recounting of it as trajectories through a semantic embedding space, then segment each trajectory into events. We applied our approach to data collected as participants watched a television episode, verbally recounted it immediately, and recounted it again one week later. Participants' delayed recountings resembled their own immediate recountings more closely than they resembled the episode itself. Their structure was specific enough to identify the immediate recounting that preceded them. Delayed recountings were also well described as geometrically simplified versions of immediate recountings, in that the events whose removal least distorts a trajectory's shape were those most likely to be dropped. What we retrieve after a delay therefore appears to be a person-specific distillation, rather than the experience as it was first encountered.

# Introduction

Suppose that you attend a friend's wedding. That evening you describe it to your partner, recounting how the ceremony ran long, how the best man's toast was unexpectedly moving, and how a small child wandered onto the dance floor midway through the first song. A week later a coworker asks how the wedding was, and you tell the story again. What is that second telling a telling of? One appealing answer is that both tellings draw on the same underlying memory of the wedding, and that whatever differences arise between them reflect the ordinary imprecision of putting an experience into words. A second possibility, which motivates the work we report here, is that your first telling has itself become part of what you are remembering.

What happens to an experience between living it and describing it later? Since our brains do not maintain an exhaustive account of everything we encounter, the very act of perceiving an experience already necessarily entails compressing and distorting it. We attend closely to some aspects of a scene while ignoring others, and we tend to interpret ambiguous moments in light of what has happened previously and we expect will happen next. But once an experience has ended, neither the experience itself nor our moment-to-moment perception of it remains available, to us or to anyone else. What remains is instead the product of a further distillation, in which our perception of the experience is reduced to something our memory systems can store and later reconstruct. Which aspects of an experience survive this second distillation may depend on what we attended to, what we anticipated would matter later, what we found interesting, and the thoughts, motivations, and biases we brought with us. Because remembering entails reconstructing an experience rather than replaying it, the resulting account often preserves the general "shape" of what happened while discarding many of its particulars (Bartlett, 1932; Schacter, 2012; Schacter et al., 2011). The units over which these distillations operate appear to be events, in that people segment continuous experience into discrete episodes at moments of change, and those boundaries shape what is remembered afterwards (Baldassano et al., 2017; Radvansky and Zacks, 2017). In prior work we characterized this transformation geometrically by modeling an experience and a person's recounting of it as trajectories through a semantic space, and asking which properties of the experience's trajectory are preserved in the recounting (Heusser et al., 2021).

Yet memories do not remain fixed once they have been encoded, pointing to further transformations that a single recounting cannot reveal. Information may be lost over time; experiences we have after an event may color how we later remember it; similar experiences may interfere with one another; and we may simply re-evaluate what happened in light of what we have learned since. The act of retrieving a memory itself appears to be one of the forces that changes it, in that a memory can become labile when it is reactivated and must be stabilized again afterwards (Hupbach et al., 2007, 2009; Nader et al., 2000; Sederberg et al., 2011). Retrieval also changes what will be accessible later—often substantially—relative to memories that go unretrieved (Karpicke and Roediger, 2007; Roediger and Karpicke, 2006). Further, the particular way in which we recount an experience appears to matter as well, since people who are asked to describe an event for one purpose later remember it in ways that are biased toward that description (Tversky and Marsh, 2000). Memories also become more semantic and less perceptually detailed, both with the passage of time and with repeated remembering (Lifanov et al., 2021). Repeated retrieval of naturalistic events over the course of a week has been found to stabilize each person's recall (for example, which details they include) without causing different people's recalls to converge (Melega et al., 2025). Taken together, these observations suggest that each time we remember an experience, the substance of the underlying memory we are retrieving may differ from what we retrieved before.

How might we compare two recountings of the same experience when the experience and both recountings take the form of unconstrained natural language? Recent work has begun to use language models to quantify the content of what individuals say when they recount what they remember (Fenerci et al., 2025; Fitzpatrick et al., 2026; Heusser et al., 2021; Manning et al., 2022). The approach developed in our prior work embeds the description of an experience, and each individual's recounting of it, into a common high-dimensional space whose coordinates reflect the concepts under consideration, and through which the experience and recountings each unfold over time as a geometric trajectory (Fitzpatrick et al., 2026; Heusser et al., 2021; Vera et al., 2025). We then segment the experience and each recounting into events and identify which event from the experience each recall event describes, yielding sequences of events that may be compared directly across people and across delays (Baldassano et al., 2017; Chen et al., 2017; Heusser et al., 2021).

Here we apply our approach to data from an experiment in which participants viewed an episode of a television show, recounted its events immediately afterward, and then recounted them again following a week-long delay. Because we collected two recountings of the same experience from each participant, we are able to ask what participants' delayed recountings most closely resemble. If retrieval recovered the original experience in an uncorrupted form, then both recountings should be about equally similar to the experience itself. We find instead that the structure of a participant's delayed recounting is more similar to that of their own immediate recounting than to that of the original episode. It is also more similar to their own immediate recounting than to other participants' immediate recountings, to the degree that an individual's immediate recall can be identified from their delayed recall on the basis of this structure alone. What governs retrieval after a delay therefore appears to be a person-specific distillation, rather than the episode as it was first experienced.

Quantifying the semantic geometry of individuals' recalls at different points in time offers the opportunity to characterize the transformations undergone by our memories beyond an initial recounting. For example, if individuals' recountings tend to retain their idiosyncratic structures after a delay, how *do* the structures of our memories tend to change with time? There is a particular qualitative change our memories undergo as we naturally forget aspects of experience, implied by common descriptors such as “fading”, “blurring”, becoming “fuzzy”, or “decaying” (Thorndike, 1913). But the nature of this change can be difficult to characterize more concretely. Here we show that this transformation is well described as geometrically simplifying the path through semantic space an individual follows as they recount an experience, in that applying existing algorithms for removing detail from a geometric curve to participants' immediate recalls identifies the details they are likely to forget following the delay.

# Results

***[OUTLINE — placeholder structure, not final prose. Bracketed italic text marks content still to be written. No statistic below has been computed or verified; every "report [STAT]" is a hole to be filled from the cited notebook's output.]***

***[FIGURE BUDGET: 8 main-text figures maximum. The plan below uses 6, leaving 2 in reserve (one likely wanted for the RDP analysis if it needs a second panel set, one for contingency). Figure 1 (`paradigm.pdf`) is the only finished main-text figure; the other PDFs already in `paper/figures/source/` are analysis outputs that may feed panels rather than figures in their own right. Nothing else is made yet.]***

## From episode and recalls to comparable event sequences

*[Walk the reader through the pipeline, briefly and conceptually (details live in the Methods), up to the point where we have three things in hand: (a) the sequence of events comprising the episode, (b) each participant's sequences of immediate and delayed recall events, and (c) the mapping between each recall event and the episode event it describes. This is the setup that everything below depends on — the point to land is that the episode and both recalls now live in a common space and are segmented into comparable units.]*

*[**Figure 1 — paradigm.** `paradigm.pdf`, already in `paper/figures/source/`. Session-1 row: instructions, episode (24 min), immediate recall (10–60 min), prediction (5 min); one-week gap; session-2 row: instructions, delayed recall (10–60 min), session-2 episode (20/22 min), recall of that episode (10–60 min). Planned revision: reduce the opacity of the blocks for the phases we don't analyze — the session-1 prediction block, and the session-2 episode-viewing and subsequent-recall blocks — so the reader's eye lands on the immediate/delayed recall pair that the paper is actually about.]*

*[**Figure 2 — methods overview.** To be made. Annotations and recall text → trajectories → event-segmentation procedure → event sequences. Loosely modeled on Fig. 2 of Heusser et al. (2021), extended with the (new) event-segmentation step. Layout still open. Candidate source material already generated: `episode-corrmat.pdf` and `k-optimization-episode.pdf` (from `event-segmentation.ipynb`), and the per-recall similarity heatmaps and match matrices `atlep1-` and `delayed-recall-similarity-heatmaps.pdf`*

*/`-matchmats.pdf`.]*

## Participants recall the episode's events in a similar order before and after the delay

*[Motivation: we wanted to know whether participants remembered the episode similarly before and after the delay. The simplest way to ask this is whether they recalled the same events in the same order.]*

*[Result: recall-order similarity. Computed in* `self-similarity-analyses.ipynb`*, section "Is delayed recall order more similar to own immediate recall order than to others' immediate recall orders?". The measure compares pairwise precedence relations over the episode events shared between two recalls, then percentile-ranks each participant's delayed-vs-own-immediate similarity against their delayed-vs-others'-immediate similarities, tested against chance (0.5) with a Wilcoxon signed-rank test. Report [STAT].]*

*[**Figure 3 — recall-order similarity.** To be made; multi-panel.]*

*[Panel A: illustration of the method. For one example participant, show the matrix of pairwise recall-order-precedence values over pairs of episode events — probably just the lower triangle — for their delayed recall beside their own immediate recall. Grey out (or reduce the opacity of) cells for event pairs not recalled in both sessions, since the comparison is restricted to pairs present in both. Then show the same thing again for that participant's delayed recall against a* different *participant's immediate recall, which is the contrast the analysis actually tests. This maps directly onto* `pairwise_precedence()` *and the* `shared_events` *argument to* `order_similarity()` *in the notebook, so the panel and the code should be checked against each other.]*

*[Panels B–D: de-slopified versions of panels B–D of the reference mock-up. Include the analogues of that mock-up's B and D so that the same panel types can reappear in the recall-structure figure (Figure 5), letting the reader compare order and structure on identical terms — this parallelism is the reason for the pairing and should be preserved in the final layout.]*

*[Note on ordering: this is the least novel analysis of the three and the one least dependent on our framework — order could be measured without any embedding model. It goes first precisely because it establishes the basic effect on the simplest possible terms, setting up the contrast with what follows.]*

## The semantic structure of participants' recalls is preserved across the delay

*[Motivation: order is only the coarsest description of a recall. Because our approach embeds recalls in a continuous semantic space, we can ask a stronger question — whether the* structure *of how a participant moved through the episode's content is preserved, not merely the sequence in which they hit the events.]*

*[**Figure 4 (optional) — 2D trajectories.** Decision deferred, and it depends on how this section's narrative settles. Argument for including it: it introduces the idea of geometric comparison and establishes the 2D paths* before *they get used as examples in Figure 5, so the reader isn't meeting the representation and the result at the same time. Argument against: it spends one of 8 figure slots on setup. Middle option — a compact version showing only the episode path plus immediate and delayed paths for a handful of participants, with the full version relegated to the supplement. Source:* `2d-trajectories.ipynb` *(*`atlep1-paths2d.pdf` *and* `delayed-paths2d.pdf`*,* `imm-del-2d.pdf`*); note those* `savefig` *calls are currently commented out in the notebook.]*

*[Result: trajectory/event-structure similarity. Computed in* `self-similarity-analyses.ipynb`*, sections "Is delayed recall more similar to own immediate recall than to episode?" and "Is delayed recall more similar to own immediate recall than to* others' *immediate recalls?". Both use DTW warp cost (average cosine distance between matched events) over mean-centered event sequences; the second percentile-ranks self against others and tests against chance. Report [STAT] for each.]*

*[Framing: this result appears strong enough to present as a "fingerprint" — i.e., that one can identify which immediate recall belongs to a given participant from the structure of their delayed recall alone. Frame it that way in both the text and the figure, but confirm the effect size supports the claim before committing to that language.]*

*[**Figure 5 — recall-structure similarity.** To be made; the largest figure in the paper, covering both analysis (A) "delayed is more similar to immediate than to the episode" and analysis (B) "delayed is more similar to own immediate than to others' immediates". Layout still open.]*

*[Example trajectory panels (3, from the current draft): the left panel is common to both analyses, so it should sit centrally in the final arrangement; the right panel is the contrast for analysis (A) and the middle panel is the contrast for analysis (B). Planned refinements: thicker lines; faint connectors between*

*DTW-matched events within each panel (already present in the draft); possibly PCHIP splines rather than straight segments between points, to make the paths read as curves. Under consideration — a fourth panel showing pairwise mean point-by-point distances between trajectories, which would require upsampling each trajectory to a common number of points and comparing them in order (or substituting DTW cost, if that turns out to read better).]*

*[Analysis (A) panel: per-participant connected dots showing each participant's delayed-to-immediate versus delayed-to-episode alignment cost. Probably without the KDEs from the draft version.]*

*[Analysis (B) panels (3): de-slopified versions of the current draft. Two of them are the direct analogues of Figure 3's panels B and D and show stronger effects — keeping the panel types visually identical across the two figures is what makes "structure beats order" legible without the reader doing arithmetic. The third gets at how person-specific the similarity is, and is the panel that carries the "fingerprint" claim.]*

*[Candidate supporting material:* `atlep1-recall-corrmats.pdf` *and* `delayed-recall-corrmats.pdf` *(from* `event-segmentation.ipynb`*).]*

## Delayed recalls are simplified versions of participants' immediate recalls

*[Motivation: having established that a participant's delayed recall structure resembles their own immediate recall structure, the natural next question is how it* changes *over the delay.]*

*[Result: trajectory simplification. Computed in* `rdp-analyses.ipynb`*. Each participant's immediate recall trajectory is simplified with the Ramer-Douglas-Peucker algorithm over a range of tolerances ($\epsilon$), and the $\epsilon$ minimizing DTW warp cost to their delayed recall trajectory is selected. Sub-results to report: (i) the relationship between how many events a participant dropped across the delay and how many RDP removes at the best-fitting $\epsilon$ (Kendall's $\tau$); (ii) the same against $\epsilon$ itself; (iii) whether RDP predicts* which *events were forgotten, benchmarked against a hypergeometric null and compared to predictions derived from other participants' trajectories. Report [STAT] for each.]*

*[**Figure 6 — RDP simplification.** To be made, and how best to* illustrate *the simplification is genuinely unresolved — flagged as an open design question rather than a layout still to be drawn. The regression plots for sub-results (i) and (ii) exist in* `rdp-analyses.ipynb` *but have no* `savefig` *calls, so nothing is exported to* `paper/figures/source` *yet. One thing worth trying: since Figure 5 will already have taught the reader*

*to read 2D trajectories, showing one participant's immediate path beside its RDP-simplified version and their actual delayed path would make "the delayed recall looks like a simplified version of the immediate one" visible directly, with the quantitative panels as support. If the analysis needs a second panel set, one of the 2 reserve figure slots can absorb it.]*

*[**@paxton**: does the simplification story want a fourth subsection on what the dropped events have in common (e.g. are they lower-salience, more peripheral to the narrative), or is that a Discussion point?]*

# Discussion

We asked participants to watch an episode of a television show, to recount what they remembered immediately afterwards, and to recount the episode again following a week-long delay. Embedding the episode's annotations and each recounting into a common semantic space allowed us to represent all three as trajectories, to segment each trajectory into events, and to compare them on common terms. Participants recalled the episode's events in a similar order before and after the delay. The semantic structure of a participant's delayed recall was more similar to the structure of their own immediate recall than to that of the episode, and it was more similar to their own immediate recall than to other participants' immediate recalls. Delayed recalls were also well described as simplified versions of immediate recalls, in that removing detail from a participant's immediate recall trajectory brought it closer to the delayed recall trajectory that followed it.

Taken together, these results suggest a way of thinking about what recounting an experience does to it. If experiences and memories may be described as trajectories through a semantic space, then a recounting is a traversal of one of those trajectories (Heusser et al., 2021). Semantic space differs from physical space in that nothing prevents a narrator from jumping between distant locations, and yet participants' recountings turn out to be strongly autocorrelated in practice, which is the regularity that our event-segmentation approach relies on. Because movement through the space is relatively continuous, describing how one arrives at a later event from an earlier one implies that the intervening region was traversed in some sense, whether it was articulated explicitly or left for the listener to fill in. Some events therefore behave like control points, in that removing one substantively changes the shape of the path. Others behave like waypoints, which fall close to the line between their neighbors and may be removed with little consequence, since travelling between the surrounding control points already implies passing near them. Trajectory simplification provides a way of telling the two apart, and this distinction appears to hold at both of the transitions we can examine. The events that simplification removes from the episode's own trajectory are the events that the fewest participants recounted immediately afterwards (**[VERIFY THIS]**), and the events that simplification removes from a participant's immediate recall trajectory are the ones that participant tended to drop across the delay.

That not all events are equally memorable is well established. Gist and central details, for example, survive delays that perceptual and peripheral details do not (Gilboa and Moscovitch, 2021; Heusser et al., 2021; Sekeres et al., 2016, 2018), and autobiographical narratives shed some classes of content faster than others (Diamond and Levine, 2020; Su et al., 2026). Memories also become more semantic and less perceptually detailed with the passage of time and with repeated remembering (Lifanov et al., 2021), and events with stronger connections to the events around them are better remembered than more isolated ones (Antony et al., 2024; Lee and Chen, 2022). What our results add is a change in where that asymmetry comes from. Centrality is ordinarily established from the outside, by asking raters which details matter to the plot, or by sorting content into categories fixed in advance. Here it follows from the geometry of an individual's own trajectory, and it is computable from that trajectory alone. An event is central because of where it sits relative to the rest of the path, and not because a rater judged it to be important. This turns the observation that some events are more memorable than others into a candidate explanation of why.

Why should waypoints be the events that go? One possibility is that they are redundant, in the sense that a listener, or the rememberer's own future self, can recover them from the events on either side. Another is that they are simply weaker in themselves, for example less distinctive, less emotionally charged, or less causally connected to the rest of the narrative (Antony et al., 2024). These accounts make similar predictions about which events will be forgotten, but they differ in what those predictions should depend on. If events are dropped because they are intrinsically weak, then a simplification derived from any participant's trajectory should predict forgetting about as well as one derived from the participant's own, since weak events would be weak for everyone. If they are dropped because they are recoverable from their neighbors, then a participant's own trajectory should do better, since control-point status is defined relative to the rest of that person's path. Our results favor the latter account. Read this way, dropping a waypoint is an economy rather than a loss, and it is worth noting that events which go unreported are not always unavailable, since a shift in perspective at retrieval can recover details that an earlier recounting left out (Anderson and Pichert, 1978; Oedekoven et al., 2017).

The same argument implies that what counts as a control point should be idiosyncratic. An event's status depends on the path it sits within, and different people who watched the same episode construct different paths through it, so the events whose removal would deform one person's trajectory need not be the events that matter for another's. This is what we take the specificity of the delayed recalls to mean. That a delayed recall can be matched to the immediate recall which preceded it, on the basis of structure alone, is a fingerprint in the sense that the term is used for the neural signatures by which individuals may be identified from functional connectivity or from their responses to naturalistic stimuli (Anderson et al., 2020; Finn et al., 2020, 2015). The parallel lies in the identification logic rather than in the measurement, since ours is a behavioral signature, recovered from what a participant said rather than from how their brain responded. People differ, for example, in where they place event boundaries in a continuous experience (Sava-Segal et al., 2023), in how much of their recall structure they share with others (Chen et al., 2017; Nguyen et al., 2019), and in the character of their autobiographical remembering more generally (Palombo et al., 2018). Repeated retrieval of naturalistic events has been found to stabilize each person's recall without causing different people's recalls to converge (Melega et al., 2025), which is what one would expect if each person is stabilizing their own set of control points.

Everything we measured came from a verbal recounting, and so a reasonable objection is that control points and waypoints are properties of storytelling rather than of memory. We think that dichotomy is less clean than it first appears. Recounting is the only observable available in a study of this kind, so the two are confounded by construction, and no analysis of transcripts alone can fully separate them. More to the point, the way a person tells a story is not merely a window onto what they remember, but also an input to what they will remember next. Retelling an event for a particular audience or purpose, for example, biases how it is recalled later (Marsh, 2007; Tversky and Marsh, 2000), and reproducing a narrative repeatedly reshapes it in ways that persist (Bergman and Roediger, 1999; St Jacques and Schacter, 2013). If the function of recounting is to let a listener, or one's own future self, re-traverse the trajectory of the original experience, then communicative efficiency and mnemonic efficiency are not competing explanations so much as the same pressure seen from two sides.

We acknowledge several limitations in the current study. First, every participant recalled the episode immediately, and so we cannot separate two accounts of what we observed. A delayed recall may resemble the immediate recall because the immediate recall was itself re-encoded and then retrieved a week later, or because a single stable memory was sampled twice and each sampling was shaped by the same person-specific distillation. Our measures are symmetric and cannot establish direction, and the design lacks an arm in which the immediate recall is withheld. Designs that do include such a control, in which retrieved information is later found nearer to what was retrieved than to what was originally presented, offer the strongest evidence that retrieval writes its own output back into the trace (Bridge and Paller, 2012; Nader and Hardt, 2009; Nader et al., 2000). Second, we used one episode, one delay, and one intervening recounting, and it remains to be seen how the picture changes across stimuli, across longer intervals, or across many successive retellings (Chan et al., 2009; Lifanov et al., 2021). Third, control-point status is defined with respect to one particular simplification algorithm, and a different algorithm might well partition the same events differently. Fourth, our procedure matches every recall event to an event in the episode, so content that a participant introduced but that does not appear in the episode falls outside what we are able to measure. Fifth, our measure of what connects one event to another is built from semantic similarity, and so it cannot on its own distinguish events that are close in meaning from events that are causally linked (Antony et al., 2024; Lee and Chen, 2022; Xu et al., 2024).

Speculatively, this account suggests what might become of a memory that is recounted many times. If each retelling preserves control points and sheds waypoints, and if each retelling then becomes the trajectory that the next one simplifies, a memory might converge over many retellings toward a skeleton of control points, with the intervening detail increasingly implied rather than stated. Whether that convergence has a floor, and whether the surviving skeleton continues to resemble the original experience or drifts steadily away from it, are questions our data cannot answer. The possibility seems worth taking seriously outside the laboratory, since the settings in which people are asked to recount an experience repeatedly are often the settings in which accuracy matters most. An eyewitness who has already described what they saw, for example, is describing a description the next time they are asked (Chan et al., 2009). A patient recounting a formative episode across successive sessions may be reshaping it in the process, and a teacher who explains an idea the same way each term may be simplifying it past the point where it still carries what made it worth explaining.

Ultimately, our work points toward a set of questions about how the geometry of a recounting relates to the experience that it describes. Do the events that behave like control points in a recounting correspond to the boundaries at which neural activity shifts during the experience itself (Baldassano et al., 2017; Chen et al., 2017)? Can they be predicted in advance from the stimulus, or are they recoverable only after the fact, from a particular person's path? Does a listener who hears a recounting actually re-traverse the speaker's trajectory, filling in the waypoints that the speaker left out? Would causal and importance judgments, collected for this same episode, show that geometric control points and causally pivotal events are one notion arrived at by different routes? Questions of this kind become tractable once what a person retains from an experience is treated as a shape that can be measured, rather than as a quantity that can only be counted. Because that shape may be recovered from what someone says, and because control points are identifiable from a single recounting, it may eventually be possible to ask what a student will still carry from a lecture, or what a patient will still carry from a conversation with their physician, before the delay that would reveal it has passed. We hope that this offers a way of thinking about forgetting as something other than loss, in that what falls away between one telling and the next is largely what the remaining structure already implies, and what survives is the shape in which an experience can be carried forward and handed on.

# Methods

Our experimental protocol was approved by the Committee for the Protection of Human Subjects at Dartmouth College. We obtained written informed consent from all participants prior to the start of both experiment sessions.

## Participants

We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received optional course credit for enrolling and were pre-screened to ensure they had not previously seen any episode of either TV show viewed during the experiment. No statistical method was used to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall phase, and 7 did not return for the second experiment session. The data we analyzed comprised the remaining 53 participants.

At the start of the experiment, we asked each participant to complete a demographic survey that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported their sex as female and 12 participants reported their sex as male. Participants reported their races as White (34 participants), Asian (16 participants), Black or African American (3 participants), American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some participants selected multiple racial categories.) A total of 52 participants reported their ethnicity as "Not Hispanic or Latino" and 1 participant reported their ethnicity as "Hispanic or Latino."

We also asked participants their native spoken language; whether they had any hearing, speech, or color vision impairments; and about any current medications or recent injuries that could impact attention or memory. All participants reported that their native spoken language was English, that they had no hearing or speech impairments, and that they had normal color vision. A total of 52 participants reported no attention- or memory-relevant medications or injuries. One participant reported sustaining a concussion more than 3 years prior to the experiment and was not excluded from our dataset.

Additionally, before each of the two experiment sessions, we asked participants about their sleep, coffee consumption, and level of alertness. Participants reported having slept between 5 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported feeling "Very sluggish," 14 reported feeling "A little sluggish," 18 reported feeling "Neutral," 15 reported feeling "A little alert," and 5 reported feeling "Very alert." At the start of session 2, 3 participants reported feeling "Very sluggish," 9 reported feeling "A little sluggish," 23 reported feeling "Neutral," 17 reported feeling "A little alert," and 3 reported feeling "Very alert."

**@jeremy**: *do we want to report class years and majors? And/or do we want to report post-experiment survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you think you recalled it, how well do you think you learned characters' names)?*

## Experiment

The experiment took place over two sessions lasting approximately one hour each, separated by 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment, participants were assigned to one of two groups (*A* or *B*), alternating based on enrollment order. During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* ("The Big Bang"; duration: 24 min 15 s). Immediately after viewing, participants were given up to one hour to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017), participants were instructed to describe the episode in as much detail as possible and to try to speak for a minimum of 10 minutes. They were told to try to recount the episode's events in their original order, but that they were free to go back and describe earlier events should they realize they had missed something. After the immediate recall phase, participants were then given up to 5 minutes to predict what they thought would happen in the show's next episode.

Upon returning for session 2, all participants were again given up to one hour to recount the *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants in Group *A* then viewed the second episode of *Atlanta* ("Streets on Lock"; duration: 21 min 57 s) while participants in Group *B* instead viewed the pilot episode of the FOX series *Arrested Development* ("Pilot"; duration: 20 min 37 s). Both groups were then given up to one hour to recount the just-watched episode in the same manner and given the same instructions as in session 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording verbal recall audio (Ziman et al., 2018). Here we analyze participants' immediate and delayed recalls of the *Atlanta* episode viewed in session 1.

## Stimulus annotation and recall transcription

To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annotations of the on-screen content during each individual camera shot throughout the episode (mean annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001) to record, for each camera shot, a set of semantically relevant features. These included a brief description of physical occurrences and characters' explicit actions, a brief description of characters' internal states (e.g., emotions, motivations) and other implicit content, the names of all characters appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the shot's setting, whether the shot took place indoors or outdoors, and whether or not there was music playing during the shot. This feature set was adapted from a similar annotation corpus created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in prior work (Heusser et al., 2021).

We then transcribed each participant's immediate and delayed verbal recall of the *Atlanta* episode. Our transcription process consisted of three steps: an initial manual transcription, a two-pass prose-formatting conversion with a large language model, and a final manual cleanup and correction step. First, we manually transcribed each recall audio file collected during the experiment. These initial transcripts reflected participants' spoken recalls verbatim, without any punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other artifacts common to unrehearsed speech. However, since transformer models (such as we used to obtain text embeddings of participants' recalls) are trained almost exclusively on written text, structural conventions like sentence breaks, punctuation, and capitalization provide an important signal these models leverage to contextual other tokens (e.g., words) and construct a representation of the text's overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions and filler words ("like", "um", etc.) that are comparatively rare in written text contribute out-of-distribution noise that can be expected to degrade embedding fidelity, and prior work has shown that automated removal of these phenomena improves performance on downstream tasks that assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants' recall transcripts to a structure and style more typical of written text. We used the `langchain_dartmouth` Python package (Stone et al., 2025) to implement this conversion as two sequential passes over each transcript. In the first pass, we prompted the model to split the transcript into sentences at natural breaking points, add punctuation including quotes around reported speech, remove clear non-word disfluencies ("um," "uh,", "hmm," etc.), collapse stuttered verbatim repetitions, and mark self-corrections and restarts with em-dashes. In the second pass, we prompted the model to remove all instances of "like" used as a filler word. We removed filler "like" instances in a separate pass because of their high frequency of occurrence in the transcripts and the additional effort to required to distinguish them from meaningful uses of "like." The full texts of our prompts for both passes can be found in *Supplementary methods*. Finally, we manually reviewed all LLM-formatted transcripts to identify and correct any errors, ensure sentence breaks were appropriate and consistent across participants, confirm that no content was improperly added to or removed from the transcripts, and resolve any ambiguities based on the prosody of participants' original recall audio.

# Analysis

## Statistics

Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches own immediate *better than* the episode, others' immediates, etc.).

## Constructing episode and recall trajectories

We adapted a method we developed in prior work (Heusser et al., 2021) to transform the *Atlanta* episode and participant's recalls of it into trajectories through a common representation space using a text embedding model. Briefly, text embedding models take in natural language text of an arbitrary length and output a fixed-length vector representation that captures the text's semantic meaning. Intuitively, an output vector of length $n$ can be viewed as a coordinate in an $n$-dimensional text embedding space. While specific model architectures and training paradigms vary, modern text embedding models are broadly trained to output similar coordinate vectors (in terms of cosine similarity, dot product, inverse Euclidean distance, or other geometric measures) for semantically similar input texts, even when those texts share few or no specific words. We obtained text embeddings for the episode and recalls using EmbeddingGemma (Vera et al., 2025), a pre-trained 308-million-parameter encoder-only transformer model that outputs coordinate vectors in a 768-dimensional embedding space.

To construct the trajectory for the *Atlanta* episode, we first converted our tabular annotations of the content present during each camera shot to a plain-text format. Separately for each shot, we concatenated the annotation text describing explicit on-screen occurrences and characters' implicit internal states. We then appended "Speech: " followed by the shot's dialogue, "Character speaking: " followed by the name of the speaker, and "Setting: " followed by the shot's setting. If any annotated fields were empty for a given shot (e.g., if no character was speaking at that time), we omitted them and their prefixes from the concatenated text. Next, we used EmbeddingGemma to transform the text snippet for each camera shot into a text embedding vector, resulting in a number-of-shots (478) by number-of-features (768) matrix. Since the camera shots varied in duration, we then resampled this series of embedding vectors to a consistent temporal resolution. We assigned each vector to the timepoint midway between the start and end of its corresponding camera shot (in seconds), and used linear interpolation to estimate a vector for each second of the episode. This yielded a number-of-seconds (1,455) by number-of-features (768) matrix—i.e., a 768-dimensional *trajectory*.

We then created similar trajectories for each participant's immediate and delayed recall of the episode. We began by splitting each recall transcript into a list of sentences, with the constraint that splits could not occur within reported speech (i.e., if a participant recounted a multi-sentence quote by a character in the episode, we treated the full quote as a single sentence-unit). We then parsed these sentences into overlapping sliding windows that spanned up to 5 consecutive sentences each. These windows grew to and shrank from their full size at the beginning and end of a participant's recall sequence—in other words, the first window contained only the first sentence of the participant's recall, the second contained the first two sentences, and so on. This ensured that all sentences from a participant's recall transcript appeared an equal number of times (5) in their sequence of sliding windows. Finally, we used EmbeddingGemma to transform each sliding window into a text embedding vector. Repeated for all recall transcripts, this yielded a separate trajectory characterizing each participant's recall from each of the two experiment sessions. The number of samples in participants' recall trajectories (i.e., the number of sliding windows parsed from their recall transcripts) ranged from 61 to 308 (mean: 136.68; standard deviation: 49.70).

**Relating episode and recall embedding vectors**

In prior work using our trajectory-analysis framework (e.g., Fitzpatrick et al., 2026; Heusser et al., 2021; Manning, 2021; Manning et al., 2022; Raccah et al., 2024), embeddings of the stimulus and participants' responses were constructed using a topic model (Latent Dirichlet Allocation; Blei et al., 2003) trained solely on the stimulus content. Here we elected to instead use a pre-trained transformer-based text embedding model due to the richer and more nuanced textual information these models are able to capture. (For example, these context-sensitive models can disambiguate the texts "Earn looked at Alfred" and "Alfred looked at Earn" in their representations, whereas bag-of-words models like LDA would represent the two identically.) However, these richer representations also introduce certain geometric nuisances we needed to account for in order to effectively relate the contents of participants' recalls to that of the episode they viewed. First, because modern transformer models are pre-trained on large and diverse text corpora, the representational capacities of their embedding spaces span an enormous breadth of semantic content. Consequently, in preliminary analyses we found that the embeddings of our annotations and participants' recalls (which were all "about" the events, themes, and characters of a single *Atlanta* episode) were strongly anisotropic—i.e., they all occupied the same relatively narrow angular region of the model's overall embedding space. Additionally, because transformer models are sensitive to the syntactic structure of text, their embedding representations carry latent information about the text's linguistic register in addition to its semantic content. Accordingly, we found there was a consistent angular offset between embeddings of the episode annotations and those of participants' recalls, reflecting the fact that the annotations comprise grammatically well-structured written text while the recalls comprise transcripts of casual unrehearsed speech.

We corrected for these representational artifacts by mean-centering the episode and recall trajectories before performing analyses that involved measuring similarities between embedding vectors within or across trajectories. Conceptually, mean-centering a given set of embedding vectors (i.e., subtracting the across-vector average from each individual vector) serves to remove representations that are common across all vectors under consideration, and instead highlight representations that *vary* between them. For example, consider the sequence of embedding vectors comprising the trajectory of the *Atlanta* episode. Common across vectors within this sequence will be latent signals driven by how we chose to format the annotation text passed to the embedding model, the annotator's particular idiolect (i.e., idiosyncratic vocabulary and writing style), and any high-level overarching themes of the episode. *Not* common across these vectors are the finer-grained semantic representations that distinguish different events, scenes, and actions throughout the episode from each other. We found that removing these non-discriminative signals (to the extent they behave as additive offsets) via mean-centering helped reduce the embeddings' anisotropy (i.e., increased the range of measured between-vector similarities) and improved our ability to segment the episode trajectory into meaningful events. Similarly, mean-centering both the episode and recalls before relating their embedding representations helped mitigate broad-scale representational differences driven by their diverging registers, and enabled us to better relate them based on event-specific, content-driven patterns.

Our approach to mean-centering the recalls was guided by the particular latent representation we wanted to remove versus retain. In segmenting each participant's recall trajectory from each session into events (see *Event-segmentation*), our goal was to discriminate between embedding vectors strictly within a single recall sequence in order to identify temporally transient content-related patterns. We therefore centered each recall trajectory (in addition to the episode) on its own centroid to remove non-discriminative representations (e.g., register, broad-scale thematic content) and maximize the contrast between vectors within that single sequence. However, the analyses shown in Figures **REF TO DTW FIG** and **REF TO RDP FIG** entailed comparing sequences of recall events not only to the episode, but also to each other, both within- and across-participants, to examine whether within-participant similarities tended to exceed across-participant similarities. Here our goal was to remove representational artifacts that could work to bias our results towards significance. Specifically, participants' idiolects (i.e., particular speaking styles and sets of words with which they desecribe the episode) will influence the embedding representations of their recalls in ways that tend to be similar between their immediate and delayed recalls, but different across participants. To correct for this confound, we jointly centered each participant's set of immediate and delayed recall events on the two sequences' shared centroid (i.e., the average vector across all events from both sessions). This served to reduce the recall embeddings' anisotropy (by removing signals driven by high-level themes of the episode they recalled) and mitigate the effects of the spoken-recall register and participant-specific idiolect (both of which are shared between a participant's recalls from the two sessions) while preserving any genuine drift in the high-level content of each participant's recall over the delay (which per-session mean-centering would instead discard).

## Event-segmentation

After constructing trajectories for the episode and each participant's immediate and delayed recall, we used hidden Markov models (HMMs; Baum and Petrie, 1966; Rabiner, 1989) to segment each trajectory into a sequence of discrete events. Given a number of states $K$, an HMM models an observed time series as a progression through $K$ hidden states, each of which has a characteristic signature in the data (i.e., a pattern that timepoints belonging to that state tend to exhibit) and tends to persist across timepoints with occasional transitions rather than fluctuating freely. Because states persist, inferring the most probable state at each timepoint segments the observed time series into a sequence of events—i.e., spans of contiguous timepoints with the same most-probable state.

We segmented the episode trajectory into events using the HMM implemented in the BrainIAK toolbox (Kumar et al., 2021), which jointly learns a mean pattern for each of $K$ states and the set of transitions between them via the Baum-Welch algorithm (Baum et al., 1970), with the added constraint that each state is visited exactly once in a strict left-to-right order (Baldassano et al., 2017). To select an appropriate number of states (i.e., events) for the episode, we used an optimization procedure we developed in prior work (Heusser et al., 2021) that aims to identify the single timescale of event representations emphasized most heavily in the episode's narrative structure. We began by segmenting the (mean-centered) episode trajectory into $K$ events for each $K$ between 2 and 50, inclusive. For each candidate segmentation, we then computed the cosine similarity between each pair of timepoints assigned to the same event, and between each pair of timepoints assigned to different events. When constructing these distributions of within- and across-event similarities, we restricted comparisons to pairs of timepoints separated by no more than the duration of the longest event in the candidate segmentation, so that the within- and across-event distributions were computed over comparable temporal distances and the across-event distribution was not dominated by distant timepoint pairs that we would expect to be dissimilar regardless of event structure. We then selected the candidate segmentation that maximized the Wasserstein distance between these within- and across-event similarity distributions. We found that the maximally distinctive and internally self-similar segmentation was yielded by $K = 20$

events. **(TODO: ADD SUPP FIG SHOWING WASSERSTEIN DIST AS FUNCTION OF K)** Finally, we averaged the embedding vectors over timepoints between each pair of adjacent event boundaries to obtain a sequence of stable event-average representations for the episode.

To segment participants' recall trajectories into sequences of events, we developed a novel HMM-driven procedure that grounded the recalls directly against the content of the episode participants viewed and recounted. Essentially, rather than re-learning separate sets of state representations from each participant's recalls (which tend to be shorter, noisier approximations of the episode content), we leveraged the state representations already learned from the episode to identify transitions between the episode events a given participant was (most probably) describing at a given point in time. First, we mean-centered the sequence of event-average embedding vectors for the episode, and each participant's sequence of per-window embedding vectors from each session, separately. For each given recall trajectory, we then computed the cosine similarity between the embedding vector for each recall window and the event-average embedding vector for each of the 20 episode events. This yielded a number-of-windows by number-of-episode-events (20) similarity matrix. We then treated the episode events as the hidden states of a hidden Markov model with fixed parameters, and treated each recall window's profile of similarities to the episode events, $z$-scored across all window-event pairs, as its distribution of emission scores over states. We imposed an ergodic transition structure (i.e., we allowed transitions from any state to any other state, since participants may recall the episode events out of order, and allowed states to be re-entered multiple times, since participants may revisit earlier events in their recalls) governed by separate additive transition penalties for forward-one transitions and all other transitions. Specifically, we imposed no penalty for remaining in the current state, a small penalty (1.5 normalized emissions units) for transitioning from a given state to the next (i.e., transitioning from recalling episode event $n$ to recalling episode event $n + 1$), and a larger penalty (3.0 normalized emissions units) for all other transitions. Finally, we decoded the most probable sequence of episode-event labels using the Viterbi algorithm (Viterbi, 1967), collapsed runs of consecutive identically labeled recall windows into events, and averaged the embedding vectors over windows within each event. This procedure yielded, for each participant's recall from each session, a sequence of event-average embedding vectors and a set of labels denoting which episode event each recall event corresponded to.

### Comparing trajectory structures

We compared the geometric structures of the episode and recall event-sequences using dynamic time warping (DTW; Sakoe and Chiba, 1978; Vintsyuk, 1968). The DTW algorithm finds the optimal order-preserving alignment between two sequences of vectors with potentially different lengths or sampling rates by locally "stretching" and "compressing" them in time to maximize their overall similarity. This alignment yields a "warp path" that maps each vector in each sequence onto one or more consecutive vectors in the other sequence with the minimum total distance between matched vectors. The summed or average distance between matched vectors along the warp path (the "warp cost") is commonly taken as a measure of the two sequences' structural dissimilarity. We computed the warp cost as the average cosine distance between DTW-matched events so that the cost values would be comparable across comparisons involving event-sequences of different lengths (whereas summed distance necessarily grows with sequence length).

# Data availability

All of the data analyzed in this manuscript may be found at https://github.com/ContextLab/memory-dynamics.

# Code availability

All of the code for running our experiment and carrying out the analyses may be found at https://github.com/ContextLab/memory-dynamics.

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# Author contributions

# Competing interests

The authors declare no competing interests.